

Costs that grow with size

Low beta + momentum after spread and market impact, 2013–2026



In Europe alone the edge is gone before \$100m.

Henrik Gjernerig · Project 10 · point-in-time universes, local currency (World in USD)

Henrik Gjernerig · LinkedIn

Most backtests charge a flat cost per trade. Real costs grow with the size of the trade.

I re-costed the strategies from my factsheet series with a spread that depends on how liquid each stock is, plus the square-root market-impact law, at \$1m, \$10m, \$100m, \$1bn and \$10bn.

For the pair that held up best (low beta plus momentum, market-neutral), the net return reaches zero at about:

- US: \$1.0bn
- World: \$33m
- EU: \$26m
- UK: \$3m
- Denmark: \$16m
- Scandinavia: no positive result even at \$1m

The best-looking markets at small size, the UK and EU, fill up first: their low-beta legs sit in thinly traded stocks. Fast signals like short-term reversal are negative already at \$10m.

The cheapest fix is patience. Holding a stock until it falls well out of the top group halves turnover and lifts capacity several times, with little loss at small size.

Lesson: many factor results are small-fund results. Ask what size the backtest was run at.

Full factsheet: <https://hgjerning.github.io/now-showing/factsheets/fs16-costs-and-capacity>

#investing #quant #factorinvesting #tradingcosts #backtesting